Note on the Elastic Curve Project

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This is a note on the elastic curve project.

1 BIG PICTURE

Let $\gamma\colon \mathbb{S}^1 \to \mathbb{R}^3$ denote a closed space curve. Let $\kappa\colon \mathbb{S}^1 \to \mathbb{R}^3$ and $\tau\colon \mathbb{S}^1 \to \mathbb{R}^3$ denote the Frenet curvature and torsion of the curve. Let ds be the arclength measure defined along the curve. The length of the curve is given by

$$E_1 = \int_{\mathbb{S}^1} ds. \tag{1}$$

The total torsion is given by

$$E_2 = \int_{\mathbb{S}^1} \tau\, ds, \tag{2}$$

and the total bending energy is given by

$$E_3 = \frac{1}{2}\int_{\mathbb{S}^1} \kappa^2\, ds. \tag{3}$$

A curve γ is an 3D Euler elastic curve (elastica) if a linear combination of the gradients of E_1, E_2, E_3 is zero. One can interpret a 3D Euler elastica as a curve that minimizes E_3 with equality constraints that E_1 and E_2 are fixed. Physically, fixing E_1 means the total length is fixed, natural for inextensible material. Fixing the total torsion E_2 may look obscure at first, but by the Călugăreanu theorem, total torsion equals to the total twist of a frame. Physically fixing E_2 amounts to imagining that the curve is an infinitesimal tube with notion of frame, and closing the curve into a closed loop would naturally fix the frame at the ends. With these physical constraints, one minimizes the total bending energy.

Let $(\cdot)' = \frac{d}{ds}$ denote the derivative with respect to arclength. The gradients are

$$\nabla E_1 = -\gamma'', \tag{4}$$

$$\nabla E_2 = -\gamma' \times \gamma''' = -(\gamma' \times \gamma'')' \tag{5}$$

$$\nabla E_3 = (\gamma''' + \frac{3}{2}|\gamma''|^2\gamma')'. \tag{6}$$

Observe the condition for being Euler elastica

$$\nabla E_3 + \lambda_2 \nabla E_2 + \lambda_1 \nabla E_1 = 0 \tag{7}$$

is a total derivative condition

$$\left(-\gamma' - \lambda_2\gamma' \times \gamma'' + \gamma''' + \frac{3}{2}|\gamma''|^2\gamma'\right)' = 0 \tag{8}$$

and thus there exists a constant $a \in \mathbb{R}^3$ so that

$$-\gamma' - \lambda_2\gamma' \times \gamma'' + \gamma''' + \frac{3}{2}|\gamma''|^2\gamma' = a. \tag{9}$$

Take a cross product with tangent vector $\times\gamma'$ on both sides of the equation to obtain

$$\lambda_2\gamma' \times (\gamma' \times \gamma'') - \gamma' \times \gamma''' = a \times \gamma' \tag{10}$$

$$\implies \quad -\lambda_2\gamma'' - (\gamma' \times \gamma'')' = (a \times \gamma)'. \tag{11}$$

Here we have used $\gamma'' \perp \gamma'$ and $|\gamma'| = 1$ to get $\gamma' \times (\gamma' \times \gamma'') = -\gamma''$. This is once again a total derivative expression. Therefore, there exists another constant $b \in \mathbb{R}^3$ so that

$$-\lambda_2\gamma' - \gamma' \times \gamma'' = a \times \gamma + b. \tag{12}$$

Take another cross product $\times\gamma'$ to obtain

$$\gamma'' = (a \times \gamma + b) \times \gamma'. \tag{13}$$

This reduced Elastica equation has much smaller order of derivative as a differential equation, compared to (8). In fact, given (13), one can also work backwards to (8) by supplying suitable multiple of γ' that were eliminated when we took the cross product with γ'.

Equation (13) can be interpreted as minimizing length with some other functionals being conserved (with $a, b \in \mathbb{R}^3$ being Lagrange multipliers). One can see that $a \times \gamma + b$ is an infinitesimal 3D rigid body motion vector field evaluated along the curve. This force that would balance with length minimization force γ'' emerges if we constrain the Hamiltonians that generate rigid body motions. These constraints happen to be some notion of 3D area vector and revolving volume.

The goal is to generate elastica using the reduced equation (13) via length minimization with area and volume constraints. These functionals make sense even at the discrete setting, in contrast to higher order energies.

2 ISOPERIMETRIC CHARACTERIZATION

We want to consider isoperimetric characterizations of closed elastic curves, as length minimizing curves with constraints on their area, and volume vectors. For a closed curve γ that is parameterized by arc-length, its length is given by

$$L(\gamma) = \int_{\mathbb{S}^1} |\gamma'|\, ds, \tag{14}$$

The *energy* of a curve

$$E(\gamma) = \tfrac{1}{2}\int_{\mathbb{S}^1} |\gamma'|^2\, ds \tag{15}$$

is an alternative variational characterization of geodesics. In contrast to the length, it also constraints the curves parametrization, so that for arc-length parameterized curves γ, the Euler–Lagrange equations of the two coincide.

For closed curves, minimizing these functionals on their own yields degenerate solutions. For the isoperimetric characterization of elastic curves, we consider constraints of a curve's area vector

$$A(\gamma) = \int_{\mathbb{S}^1} \gamma \times \gamma'\, ds, \tag{16}$$

and the volume vector

$$V(\gamma) = \int_{\mathbb{S}^1} \gamma \times (\gamma \times \gamma')\, ds. \tag{17}$$

One can check that for an arc-length parameterized curve, the variational gradients of these fucntionals are given by

$$dL(\mathring{\gamma}) = -\int_{\mathbb{S}^1} \langle \gamma'', \mathring{\gamma} \rangle, \tag{18}$$

$$\langle a, dA(\mathring{\gamma}) \rangle = \int_{\mathbb{S}^1} \langle 2\gamma' \times a, \mathring{\gamma} \rangle, \tag{19}$$

$$\langle b, dV(\mathring{\gamma}) \rangle = \int_{\mathbb{S}^1} \langle \gamma' \times (b \times \gamma), \mathring{\gamma} \rangle \tag{20}$$

for any $a, b \in \mathbb{R}^3$.

Define the energy

$$\tilde{E}(\gamma) = L(\gamma) + \langle a, A(\gamma) \rangle + \langle b, V(\gamma) \rangle, \tag{21}$$

then its variational gradient is given by

$$d\tilde{E}(\mathring{\gamma}) = \int_{\mathbb{S}^1} \langle -\gamma'' + 2\gamma' \times a + \gamma' \times (b \times \gamma), \mathring{\gamma} \rangle. \tag{22}$$

Hence, the Euler–Lagrange equation is given by

$$-\gamma'' + 2\gamma' \times a + \gamma' \times (b \times \gamma) = 0. \tag{23}$$

Taking a cross product $\gamma' \times$ we obtain

$$-\gamma' \times \gamma'' + 2\langle a, \gamma' \rangle \gamma' - 2a + b \times \gamma - \langle b \times \gamma, \gamma' \rangle \gamma' = 0. \tag{24}$$

Collecting all coefficients for tangential terms and denoting them by $c \in \mathbb{R}$, this gives

$$-\gamma' \times \gamma'' - 2a + b \times \gamma + c\gamma' = 0, \tag{25}$$

which is of the form of Eq. (**??**) after appropriate renaming of the constants. Hence, stationary points of $\tilde{E}$ will be elastic curves.

We should also show the converse statement.

2.1 Transformation Rules

3 DISCRETE LENGTH, ENERGY, AREA AND VOLUME VECTORS

For our numerical treatment of the optimization problem, we will need suitable discrete analogues of the variational energies and integral constraints that appear in our problem statement.

A *discrete curve* is a map $\gamma\colon \{0, \dots, n\} \to \mathbb{R}^3$. A discrte curve is said to be *parameterized by arc-length* if $|\gamma_{i+1} - \gamma_i| = |\gamma_{j+1} - \gamma_j|$ for all $i, j \in \{0, \dots, n\}$ and *closed* if $\gamma_n = \gamma_0$.

For now, we will only consider closed curves.

Analogous to the continuous case, the length of a discrete curve is given by

$$\mathsf{L}(\gamma) = \sum_{i=0}^{n-1} |\gamma_{i+1} - \gamma_i| \tag{26}$$

ans its energy is given by

$$\mathsf{E}(\gamma) = \tfrac{1}{2} \sum_{i=0}^{n-1} |\gamma_{i+1} - \gamma_i|^2. \tag{27}$$

For the constraints, we have the area

$$\mathsf{A}(\gamma) = \tfrac{1}{2} \sum_{i=0}^{n-1} \gamma_i \times \gamma_{i+1} \tag{28}$$

and volume

$$\mathsf{V}(\gamma) = \tfrac{1}{3} \sum_{i=0}^{n-1} \tfrac{\gamma_i + \gamma_{i+1}}{2} \times (\gamma_i \times \gamma_{i+1}). \tag{29}$$

4 NUMERICAL OPTIMIZATION

That is, we are interested in solving the problem of the form

$$\min_{\gamma} E(\gamma) \tag{30}$$

$$\text{s.t. } G_{A_0, V_0}(\gamma) = 0, \tag{31}$$

where

$$G(\gamma) = \begin{pmatrix} A(\gamma) - A_0 \\ V(\gamma) - V_0 \end{pmatrix} \tag{32}$$

for some fixed $A_0, V_0 \in \mathbb{R}^3$. For notational convenience, we will typically omit the subscripts of G.

Our current ansatz for solving these problems is based on the Augmented Lagrangian formulation, where we want to solve the min-max-problem

$$\min_{\gamma} \max_{\lambda} \mathcal{L}(\gamma, \lambda) \tag{33}$$

for the Augmented Lagrangian function

$$\mathcal{L}(\gamma, \lambda) = E(\gamma) - \langle \lambda | G(\gamma) \rangle. \tag{34}$$

Note that $\lambda \in \mathbb{R}^6$ absorbs the two Lagrange multipliers $a, b \in \mathbb{R}^3$ of the previous sections.

A simple gradient descent approach for the minimization procedure follows the dynamics

$$\dot{\gamma} = -M_1^{-1}(\operatorname{grad} E(\gamma) - \langle \lambda | \operatorname{grad} G(\gamma) \rangle) \tag{35}$$

$$\dot{\lambda} = -M_2^{-1} G(\gamma). \tag{36}$$

The two mass matrices M_1, M_2 can be chosen as desired. To discretize this in time, we define for any time-discrete function that

$$\Delta(\cdot)^n := (\cdot)^n - (\cdot)^{n-1}. \tag{37}$$

Our optimization is then performed as iterations of the form

$$\frac{\Delta\gamma^{n+1}}{\Delta t} = -M_1^{-1}(\operatorname{grad} E(\gamma^{n+1}) - (\lambda^n + \Delta\lambda^n) \operatorname{grad} G(\gamma^n + \Delta\gamma^n)) \tag{38}$$

$$\frac{\Delta\lambda^{n+1}}{\Delta t} = -M_2^{-1} G(\gamma^{n+1}). \tag{39}$$

Here the quantities of the form $(\cdot)^n + \Delta(\cdot)^n$ are to be interpreted as extrapolations of the quantity $(\cdot)^n$ based on the previous update.

To improve convergence rates on this simple gradient descent scheme with fixed step length, we currently add momentum (as in rolling ball) to the method, which seems to help for now. Ultimately, we should just compute Hessians as well, and use a Newton-type solver that allows for constraints (*e.g.*, some 'trust-constr' type solver) that supports proper line-search.

4.1 Gradient computation

To perform the optimization as efficient as possible, we need gradients of the discrete functionals involved. To this end, we consider perturbations $\mathring{\gamma}_j$ of individual vertices γ_j. Specifically, we define

$$\gamma_\epsilon := (0, \cdots, 0, \gamma_j + \epsilon\,\mathring{\gamma}_j, 0, \cdots, 0)$$

and, for any discrete functional F, write

$$d\mathsf{F}(\mathring{\gamma}_j) := \tfrac{d}{d\epsilon}\Big|_{\epsilon=0}\mathsf{F}(\gamma_\epsilon).$$

4.1.1 Area gradient. For perturbations of an interior vertex with index $j \in \{1,\ldots,n-1\}$, the gradient of the discrete area functional Eq. (28) is given by

$$\begin{aligned} d\mathsf{A}(\mathring{\gamma}_j) &= \tfrac{1}{2}\gamma_{j-1}\times\mathring{\gamma}_j + \tfrac{1}{2}\mathring{\gamma}_j\times\gamma_{j+1} \\ &= \tfrac{1}{2}(\gamma_{j-1}-\gamma_{j+1})\times\mathring{\gamma}_j = [\tfrac{1}{2}(\gamma_{j-1}-\gamma_{j+1})]_\times\mathring{\gamma}_j. \end{aligned}$$

Here, $[\cdot]_\times$ denotes the matrix representing the cross product on $\mathbb{R}^3$.

Note that, since for now the curves are assumed to be closed, for perturbations of boundary vertices, the same holds true, but the index shifts are to be taken $j \pm 1 \mod n$.

Therefore, the gradient of the discrete area functional is given by the $n\times 3$ matrix

$$d\mathsf{A} = \left[[\tfrac{1}{2}(\gamma_{(j-1)\%n}-\gamma_{(j+1)\%n})]_\times\right]_{j=0,\ldots,n-1}.$$

4.1.2 Volume gradient. Similarly, we compute the gradient of the discrete volume functional. For perturbations of an interior vertex with index $j \in \{1,\ldots,n-1\}$, its gradient is given by

$$\begin{aligned} d\mathsf{V}(\mathring{\gamma}_j) &= \tfrac{1}{3}\left(\tfrac{\gamma_{j-1}+\gamma_j}{2}\times(\gamma_{j-1}\times\gamma_j)\right)^\circ + \tfrac{1}{3}\left(\tfrac{\gamma_j+\gamma_{j+1}}{2}\times(\gamma_j\times\gamma_{j+1})\right)^\circ \\ &= \tfrac{1}{6}\mathring{\gamma}_j\times(\gamma_{j-1}\times\gamma_j) + \tfrac{1}{3}\tfrac{\gamma_{j-1}+\gamma_j}{2}\times(\gamma_{j-1}\times\mathring{\gamma}_j) \\ &\quad \tfrac{1}{6}\mathring{\gamma}_j\times(\gamma_j\times\gamma_{j+1}) + \tfrac{1}{3}\tfrac{\gamma_j+\gamma_{j+1}}{2}\times(\mathring{\gamma}_j+\gamma_{j+1}) \\ &= -\tfrac{1}{6}[\gamma_{j-1}\times\gamma_j]_\times\mathring{\gamma}_j + \tfrac{1}{3}[\tfrac{\gamma_{j-1}+\gamma_j}{2}]_\times[\gamma_{j-1}]_\times\mathring{\gamma}_j \\ &\quad - \tfrac{1}{6}[\gamma_j\times\gamma_{j+1}]_\times\mathring{\gamma}_j - \tfrac{1}{3}[\tfrac{\gamma_j+\gamma_{j+1}}{2}]_\times[\gamma_{j+1}]_\times\mathring{\gamma}_j. \end{aligned}$$

So again, the gradient can be written in the form of an $n\times 3$ matrix, by dropping the $\mathring{\gamma}_j$'s and wrapping around the index shifts.

5 SE(3)-ACTION AND CASIMIRS

We want to get a better understanding of how exactly the area and volume vectors parameterize the space of closed elastic curves. To this end, we consider their transformation behavior when the curve is displaced by a rigid body transformation.

Lemma 5.1. *Let γ be a (continuous or discrete) elastic curve and $\tilde{\gamma} = R\gamma + a$ for some $R \in \mathrm{SO}(3)$ and $a \in \mathbb{R}^3$. Then, their area and volume vectors related by*

$$\begin{aligned} A(\tilde{\gamma}) &= R\,A(\gamma) \\ V(\tilde{\gamma}) &= R\,V(\gamma) + a\times R\,A(\gamma), \end{aligned}$$

if γ is continuous, respectively, for the discrete case,

$$\begin{aligned} \mathsf{A}(\tilde{\gamma}) &= R\,\mathsf{A}(\gamma) \\ \mathsf{V}(\tilde{\gamma}) &= R\,\mathsf{V}(\gamma) + a\times R\,\mathsf{A}(\gamma). \end{aligned}$$